MAXWELL PIPER

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(319) 621-8929

EDUCATION

Reed College *Aug 2018 - May 2022*
Physics - BA

WORK EXPERIENCE

Observing Assistant *September 2022 - Present*
W. M. Keck Observatory, Waimea, HI

Responsible for independent nighttime operation of the Keck telescope in a high-altitude, safety-critical environment.

- Operate the Keck telescopes and instruments during nighttime observing, ensuring safe and efficient data acquisition in a high-risk, high-altitude environment.
- Identify, investigate, and respond to abnormal system behavior in real time, minimizing downtime and preventing escalation of issues.
- Maintain situational awareness of interdependent systems (telescope, dome, instruments, weather) to support continuous and efficient observing.
- Execute system recovery actions during faults, including subsystem reinitialization and server-level interventions, to restore functionality.
- Perform operational support tasks such as cryogenic servicing and hardware configuration adjustments for instrument readiness
- Make safety-critical decisions regarding weather and road conditions, including terminating observations and evacuating the summit when necessary.
- Coordinate with observers to balance scientific goals with operational and safety constraints.

Physics Teaching Assistant *August 2021 - December 2021*
Reed College, Portland, OR

Teaching Assistant for PHYS164 - Stars and Stellar Systems.

Telescope Operator *August 2018 - May 2022*
Reed College, Portland, OR

Trained operator of Reed's 12" Meade LX200 ACF Telescope and hosted observation nights open to the public.

TECHNICAL SKILLS

- Python (data analysis, pipeline development)
- Linux command line (basic system interaction)
- Astronomical data formats (FITS, MaNGA pipelines)
- Telescope and instrument operations
- Troubleshooting of software/hardware systems

RESEARCH EXPERIENCE

Senior Thesis *August 2021 - May 2022*
Reed College, Portland, OR *2 Semesters*

Investigated the characteristics of Counter-Rotating Disk galaxies using the 16th and 17th MaNGA Sloan Digital Sky Survey. Thesis advisor: Prof. Alison Crocker, Reed College.

Astrophysics Research Internship *May 2021 - August 2021*
Reed College, Portland, OR *10 Weeks*

Created aperture corrections for stellar velocity dispersion measurements for late-type galaxies using the 15th data release of the MaNGA Sloan Digital Sky Survey. Project supervisor: Prof. Alison Crocker, Reed College.

PRESENTATIONS

240th American Astronomical Society Conference (cancelled due to COVID) *January 2022*
"Stellar Velocity Dispersion Corrections for Late-Type Galaxies": Poster Presentation
Bibcode: 2022AAS...24012603P

30th Annual Murdock College Science Research Conference *November 2021*
"Stellar Velocity Dispersion Corrections for Late-Type Galaxies": Poster Presentation

Reed College Poster Fair *August 2021*
"Stellar Velocity Dispersion Corrections for Late-Type Galaxies": Poster Presentation

LEADERSHIP / ADDITIONAL EXPERIENCE

Hawaii Epic Cycling *September 2025 - Present*
Cycling Guide

Lead clients on luxury long-distance cycling routes up Mauna Kea, Mauna Loa, and around Hawaii island with the responsibility of group safety, route planning, and adapting to changing conditions.

Reed College Cycling Club *August 2021 - May 2022*
Founder

Founded the Reed College Cycling Club. The primary goal of the club is to connect Reed students that enjoy riding bikes with each other, and to foster an environment where anyone is able to explore the Portland, OR, area on a bike.